



# 11433 - Constraining Dust Formation at Cosmic Dawn with Luminous UV Bump Galaxies

Cycle: 5, Proposal Category: GO

## INVESTIGATORS

| <i>Name</i>                                | <i>Institution</i>                                       |
|--------------------------------------------|----------------------------------------------------------|
| <b>Katherine Ormerod (PI) (ESA Member)</b> | <b>Liverpool John Moores University</b>                  |
| Dr. Renske Smit (CoI) (ESA Member) (CoPI)  | Liverpool John Moores University                         |
| Dr. Joris Witstok (CoI) (ESA Member)       | Cosmic Dawn Center, Niels Bohr Institute                 |
| Prof. Michael Maseda (CoI) (US Admin CoI)  | University of Wisconsin - Madison                        |
| Dr. Anna G de Graaff (CoI)                 | Center for Astrophysics   Harvard & Smithsonian          |
| Dr. Irene Shivaei (CoI) (ESA Member)       | Centro de Astrobiologia - CAB                            |
| Dr. Francesco D'Eugenio (CoI) (ESA Member) | University of Cambridge, Kavli Institute for Cosmology   |
| Dr. Mauro Stefanon (CoI) (ESA Member)      | Universitat de Valencia                                  |
| Ms. Sian Grace Phillips (CoI) (ESA Member) | Universitat de Valencia                                  |
| Prof. Daniel P. Stark (CoI)                | University of California - Berkeley                      |
| Dr. Hanae Inami (CoI)                      | Hiroshima University                                     |
| Dr. Hiddo Algera (CoI)                     | Academia Sinica, Institute of Astronomy and Astrophysics |
| Dr. Jacqueline Hodge (CoI) (ESA Member)    | Universiteit Leiden                                      |
| Lucie Emily Rowland (CoI) (ESA Member)     | Universiteit Leiden                                      |
| Prof. Manuel Aravena (CoI)                 | Universidad Diego Portales                               |
| Prof. Paul van der Werf (CoI) (ESA Member) | Universiteit Leiden                                      |
| Rebecca Fisher (CoI) (ESA Member)          | University of Manchester                                 |
| Dr. Rychard Bouwens (CoI) (ESA Member)     | Universiteit Leiden                                      |
| Sander Schouws (CoI) (ESA Member)          | Universiteit Leiden                                      |
| Dr. Themiya Nanayakkara (CoI)              | Swinburne University of Technology                       |
| Dr. Yoshinobu Fudamoto (CoI)               | Chiba University                                         |

**OBSERVATIONS**

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i> | <i>Observing Template</i> | <i>Science Target</i> |
|--------------------|--------------------|--------------|---------------------------|-----------------------|
| Observation Folder |                    |              |                           |                       |
|                    | 1                  | GNWY         | NIRSpec IFU Spectroscopy  | (1) GNWY-737942023    |
|                    | 2                  | REBELS-08    | NIRSpec IFU Spectroscopy  | (2) REBELS-08         |

**ABSTRACT**

Dust is a fundamental component of the interstellar medium (ISM), absorbing UV and optical photons and re-emitting in the far-infrared. Some dust attenuation curves exhibit a strong feature known as the 2175Å UV bump, often attributed to carbonaceous dust production from AGB stars. Recent JWST detections of this feature out to  $z \sim 7.55$  suggest rapid dust formation inconsistent with AGB lifetimes.

We propose to target the only known galaxies at  $z > 3$  with the combination of bump strength and UV luminosity required to obtain detailed constraints on the bump shape and strength, shedding light on early dust build-up. Deep NIRSpec/IFU PRISM observations of two galaxies at  $z=6.7-7.1$  will enable accurate determination of the central wavelength and width of the bump features and will conclusively differentiate between PAHs and graphite grains as the origin of the UV bump in high-redshift attenuation curves.

Using the IFU, we will resolve the UV bump spectroscopically for the first time. Resolved star formation histories will identify the stellar populations - including the fraction of SNe and AGB stars - responsible for early dust production, while comparisons of the bump strength and sSFR will probe accelerated PAH formation.

Finally, by combining our results with existing ALMA/NOEMA dust continuum observations for the first time within the EoR, we will spatially compare small and large dust grains, while using [CII] observations to assess enhanced [CII] luminosity due to PAH-dominated photo-electric heating.

**OBSERVING DESCRIPTION**

We propose NIRSpec IFU PRISM observations of two luminous UV bump galaxies at  $z=6.7-7.1$ . We will use the integrated spectra to constrain the dust composition from the shape of the UV bump, using the peak wavelength and width of the feature to differentiate between PAHs and graphite grains as its origin. We will carry out a spatially resolved analysis of the UV bump, comparing the bump strength to properties such as star formation histories and specific star formation rates obtained through resolved SED fitting, to constrain the sources of dust grain formation (SNe vs AGB). We will test theoretical predictions regarding PAH formation, and provide the first resolved comparison of the UV bump strength to existing FIR

constraints.

Targets:

GNWY-7379420231 exhibits the strongest known UV bump at  $z > 4$ , and is the only source in which the UV bump has been resolved in NIRCам imaging at such redshifts, making this the ideal target for IFU follow up.

REBELS-08 is the most UV-luminous galaxy known to exhibit a strong UV bump feature, and in contrast to other high redshift UV bump sources, does not exhibit a shift in peak wavelength.

Observing both of these targets will enable us to determine whether the variation between sources is intrinsic, or appears due to the SNR of current data. We only request to observe two targets, as there are no other galaxies which show a strong enough UV bump and are sufficiently bright enough to achieve the required significance.

NIRSpec IFU Background, Calibration, and Dithering:

We do not request target acquisition observations as the sources do not fill the IFU FOV, therefore the pointing accuracy of the telescope is sufficient. We use the NRSIRS2 readout pattern with 12 dithers in a medium cycle dither pattern. This allows us to avoid the open shutters, while retaining good sub-pixel sampling, and leaving empty pixels free for background subtraction. We do not request dedicated background observations and will use empty pixels for background subtraction (consistent with the findings of the TEMPLATES program). We restrict the PA range of the telescope such that bright objects do not fall within the NIRSpec footprint, therefore we do not require leakcal calibrations.

In total, we request 19 hours for this proposal, including overheads.

# Proposal 11433 - Targets - Constraining Dust Formation at Cosmic Dawn with Luminous UV Bump Galaxies

| Fixed Targets | #                                                                                                          | Name           | Target Coordinates                                                                  | Targ. Coord. Corrections | Miscellaneous |
|---------------|------------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------|--------------------------|---------------|
|               | (1)                                                                                                        | GNWY-737942023 | RA: 12 37 37.9410 (189.4080875d)<br>Dec: +62 20 22.85 (62.33968d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies, Lyman-break galaxies]<br>Extended=YES |                |                                                                                     |                          |               |
| Fixed Targets | (2)                                                                                                        | REBELS-08      | RA: 02 19 35.1300 (34.8963750d)<br>Dec: -05 23 19.20 (-5.38867d)<br>Equinox: J2000  |                          |               |
|               | Comments:<br>Category=Galaxy<br>Description=[High-redshift galaxies, Lyman-break galaxies]<br>Extended=YES |                |                                                                                     |                          |               |
|               |                                                                                                            |                |                                                                                     |                          |               |

# Proposal 11433 - Observation 1 - Constraining Dust Formation at Cosmic Dawn with Luminous UV Bump Galaxies

| Observation                                                                                                       | <b>Proposal 11433, Observation 1: GNWY</b> <span style="float: right;">Fri Mar 13 21:05:55 GMT 2026</span>                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------|--------------------------|------------------|---------|--------|---------|---------------|--------------------|---------------------|-----------------|------------------|--------------------|--------------------------|------------------|---------|----------------|-------------------------------------------------------------------------------------|---------------|--------------------|---------------------|-----------------|---|-------------|---------|----|---|-------|------|------|----|----|-----------|--------|
|                                                                                                                   | <b>Diagnostic Status: Warning</b><br>Observing Template: NIRSspec IFU Spectroscopy                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| Diagnostics                                                                                                       | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
|                                                                                                                   | (Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.                                                                                                                                                                                                                                                                                          |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| Fixed Targets                                                                                                     | <table border="1"> <thead> <tr> <th>#</th><th>Name</th><th>Target Coordinates</th><th>Targ. Coord. Corrections</th><th>Miscellaneous</th></tr> </thead> <tbody> <tr> <td>(1)</td><td>GNWY-737942023</td><td>           RA: 12 37 37.9410 (189.4080875d)<br/>           Dec: +62 20 22.85 (62.33968d)<br/>           Equinox: J2000         </td><td></td><td></td></tr> </tbody> </table>                                                                                                                      |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     | #               | Name             | Target Coordinates | Targ. Coord. Corrections | Miscellaneous    | (1)     | GNWY-737942023 | RA: 12 37 37.9410 (189.4080875d)<br>Dec: +62 20 22.85 (62.33968d)<br>Equinox: J2000 |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
|                                                                                                                   | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Name                                                                                | Target Coordinates | Targ. Coord. Corrections | Miscellaneous    |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| (1)                                                                                                               | GNWY-737942023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | RA: 12 37 37.9410 (189.4080875d)<br>Dec: +62 20 22.85 (62.33968d)<br>Equinox: J2000 |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| <i>Comments:</i><br>Category=Galaxy<br>Description=[High-redshift galaxies, Lyman-break galaxies]<br>Extended=YES |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| Template                                                                                                          | <table border="1"> <thead> <tr> <th>TA Method</th><th>HFF Readout Mode</th></tr> </thead> <tbody> <tr> <td>NONE</td><td>false</td></tr> </tbody> </table>                                                                                                                                                                                                                                                                                                                                                      |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     | TA Method       | HFF Readout Mode | NONE               | false                    |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
|                                                                                                                   | TA Method                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | HFF Readout Mode                                                                    |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| NONE                                                                                                              | false                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| Dithers                                                                                                           | <table border="1"> <thead> <tr> <th>#</th><th>Dither Type</th><th>Size</th><th>Starting Point</th><th>Number of Points</th><th>Points</th></tr> </thead> <tbody> <tr> <td>1</td><td>CYCLING</td><td>MEDIUM</td><td>1</td><td>12</td><td></td></tr> </tbody> </table>                                                                                                                                                                                                                                           |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     | #               | Dither Type      | Size               | Starting Point           | Number of Points | Points  | 1              | CYCLING                                                                             | MEDIUM        | 1                  | 12                  |                 |   |             |         |    |   |       |      |      |    |    |           |        |
|                                                                                                                   | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Dither Type                                                                         | Size               | Starting Point           | Number of Points | Points  |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| 1                                                                                                                 | CYCLING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MEDIUM                                                                              | 1                  | 12                       |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| Spectral Elements                                                                                                 | <table border="1"> <thead> <tr> <th>#</th><th>Grating/Filter</th><th>Readout Pattern</th><th>Groups/Int</th><th>Integrations/Exp</th><th>Leakcal</th><th>Dither</th><th>Autocal</th><th>Total Dithers</th><th>Total Integrations</th><th>Total Exposure Time</th><th>Optional ETC ID</th></tr> </thead> <tbody> <tr> <td>1</td><td>PRISM/CLEAR</td><td>NRSIRS2</td><td>16</td><td>3</td><td>false</td><td>true</td><td>NONE</td><td>12</td><td>36</td><td>42541.203</td><td>264551</td></tr> </tbody> </table> |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     | #               | Grating/Filter   | Readout Pattern    | Groups/Int               | Integrations/Exp | Leakcal | Dither         | Autocal                                                                             | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID | 1 | PRISM/CLEAR | NRSIRS2 | 16 | 3 | false | true | NONE | 12 | 36 | 42541.203 | 264551 |
|                                                                                                                   | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Grating/Filter                                                                      | Readout Pattern    | Groups/Int               | Integrations/Exp | Leakcal | Dither | Autocal | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| 1                                                                                                                 | PRISM/CLEAR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NRSIRS2                                                                             | 16                 | 3                        | false            | true    | NONE   | 12      | 36            | 42541.203          | 264551              |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
| Special Requirements                                                                                              | Aperture PA Range 90 to 145 Degrees (V3 311.02835083 to 6.02835083)                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |
|                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                     |                    |                          |                  |         |        |         |               |                    |                     |                 |                  |                    |                          |                  |         |                |                                                                                     |               |                    |                     |                 |   |             |         |    |   |       |      |      |    |    |           |        |

# Proposal 11433 - Observation 2 - Constraining Dust Formation at Cosmic Dawn with Luminous UV Bump Galaxies

|                      |                                                                                                                                        |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                 |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|------------|------------------|----------------|--------------------------|------------------|---------------|--------------------|---------------------|-----------------|
| Observation          | <b>Proposal 11433, Observation 2: REBELS-08</b> <span style="float: right;">Fri Mar 13 21:05:55 GMT 2026</span>                        |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                 |
|                      | <b>Diagnostic Status: Warning</b><br>Observing Template: NIRSspec IFU Spectroscopy                                                     |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                 |
| Diagnostics          | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                            |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                 |
| Fixed Targets        | #                                                                                                                                      | Name           | Target Coordinates                                                                 |            |                  |                | Targ. Coord. Corrections |                  | Miscellaneous |                    |                     |                 |
|                      | (2)                                                                                                                                    | REBELS-08      | RA: 02 19 35.1300 (34.8963750d)<br>Dec: -05 23 19.20 (-5.38867d)<br>Equinox: J2000 |            |                  |                |                          |                  |               |                    |                     |                 |
| Template             | <i>Comments:</i><br><i>Category=Galaxy</i><br><i>Description=[High-redshift galaxies, Lyman-break galaxies]</i><br><i>Extended=YES</i> |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                 |
|                      | TA Method                                                                                                                              |                |                                                                                    |            |                  |                | HFF Readout Mode         |                  |               |                    |                     |                 |
|                      | NONE                                                                                                                                   |                |                                                                                    |            |                  |                | false                    |                  |               |                    |                     |                 |
| Dithers              | #                                                                                                                                      | Dither Type    |                                                                                    | Size       |                  | Starting Point |                          | Number of Points |               | Points             |                     |                 |
|                      | 1                                                                                                                                      | CYCLING        |                                                                                    | MEDIUM     |                  | 1              |                          | 12               |               |                    |                     |                 |
| Spectral Elements    | #                                                                                                                                      | Grating/Filter | Readout Pattern                                                                    | Groups/Int | Integrations/Exp | Leakcal        | Dither                   | Autocal          | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|                      | 1                                                                                                                                      | PRISM/CLEAR    | NRSIRS2                                                                            | 10         | 1                | false          | true                     | NONE             | 12            | 12                 | 8928.401            |                 |
| Special Requirements | Aperture PA Range 70 to 220 Degrees (V3 291.02835083 to 81.02835083)                                                                   |                |                                                                                    |            |                  |                |                          |                  |               |                    |                     |                 |